Supervyse OPF SMASH User Guide

Revision 1.0
02/2025

Copyright (c) 2025, All Rights Reserved. Insyde Software.

No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form, or by any means, electronic, mechanical, magnetic, optical, chemical, manual or otherwise, without the prior written permission of Insyde Software Corp.

Disclaimer

Insyde Software provides this document and the programs "as is" without warranty of any kind, either expressed or implied, including, but not limited to, the implied warranties of merchantability or fitness for a particular purpose.

This document could contain technical inaccuracies or typographical errors. Changes are periodically made to the information herein; these changes will be incorporated in future revisions of this document. Insyde Software is under no obligation to notify any person of the changes.

The following trademarks are used in this document:

Insyde and InsydeH2O are registered trademarks or trademarks of Insyde Software. All other trademarks or trade names are property of their respective holders.

Contents

Disclaimer 2

1 Introduction 5

1.1 Features 6

1.2 Flow Chart 7

1.3 How To Use 7

1.4 Examples 8

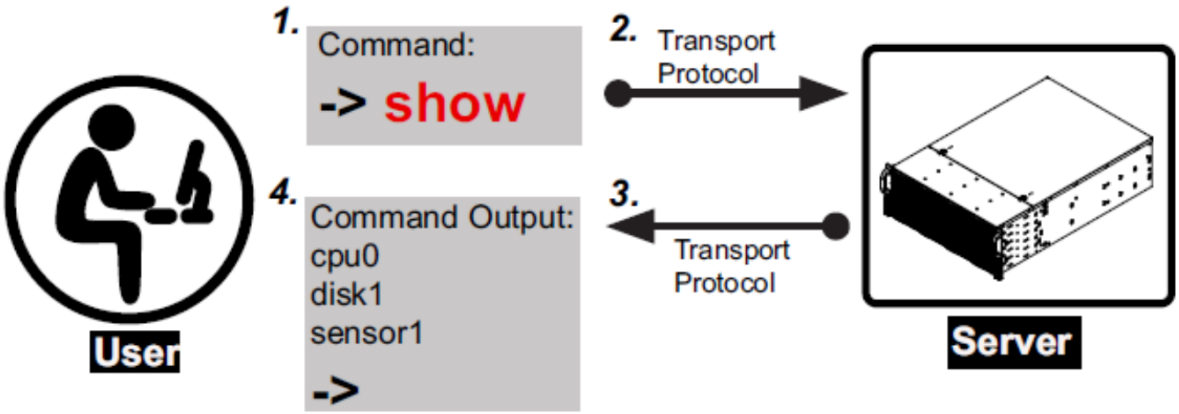
Revision History

Revision Number	Description	Author	Reviewer	Release Date
1.0	Initial release.	Robert Shih	Brad Chang	2025-02-19

1 Introduction

System Management Architecture for Server Hardware (SMASH)

The SMASH platform, developed by Distributed Management Task Force, Inc. (DMTF), delivers a host of architecture-based, Industry-standard protocols that will allow IT professionals to simplify the task of managing multiple network systems in a data center.



SMASH-CLP User Interface

Command Line Protocol (CLP)

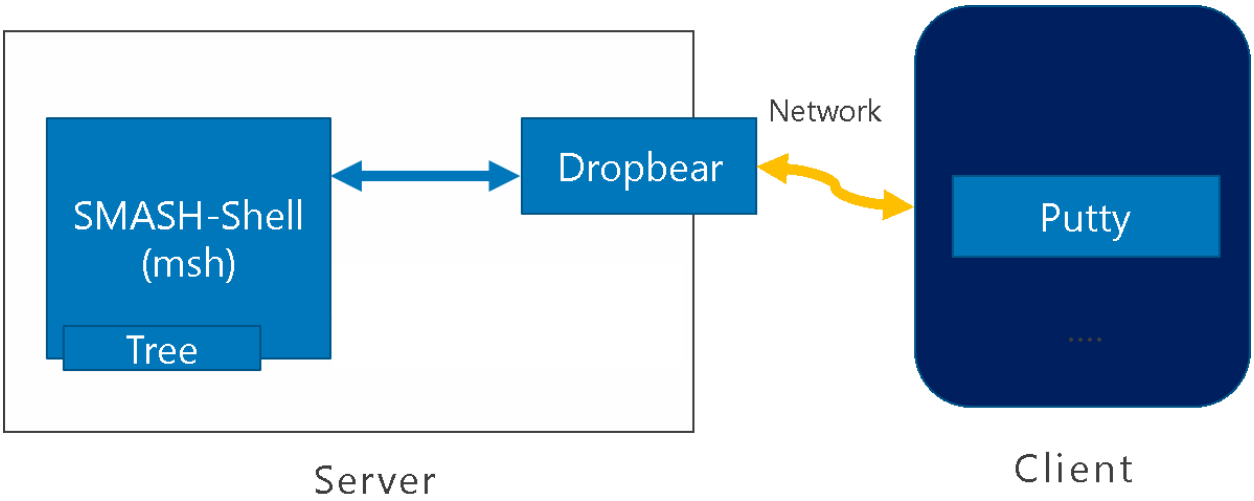
The SMASH Command Line Protocol Architecture allows the manager to access and control the devices. And the manager could use any text-console tools which use ssh connection to communicate with our server without installing any special programs. Besides, the IPMI CIM mapping had be implemented. In the other words, To power on/off server and read sensor data are all possible.

Common Information Model (CIM)

The Common Information Model (CIM) is an open standard that defines how managed elements in an IT environment are represented as a common set of objects and relationships between them. This is intended to allow consistent management of these managed elements, independent of their manufacturer or provider.

SMASH-SHELL

The open-source solution Dropbear as the SSH channel. Then implement a SMASH-shell as a total function for SMASH protocol.

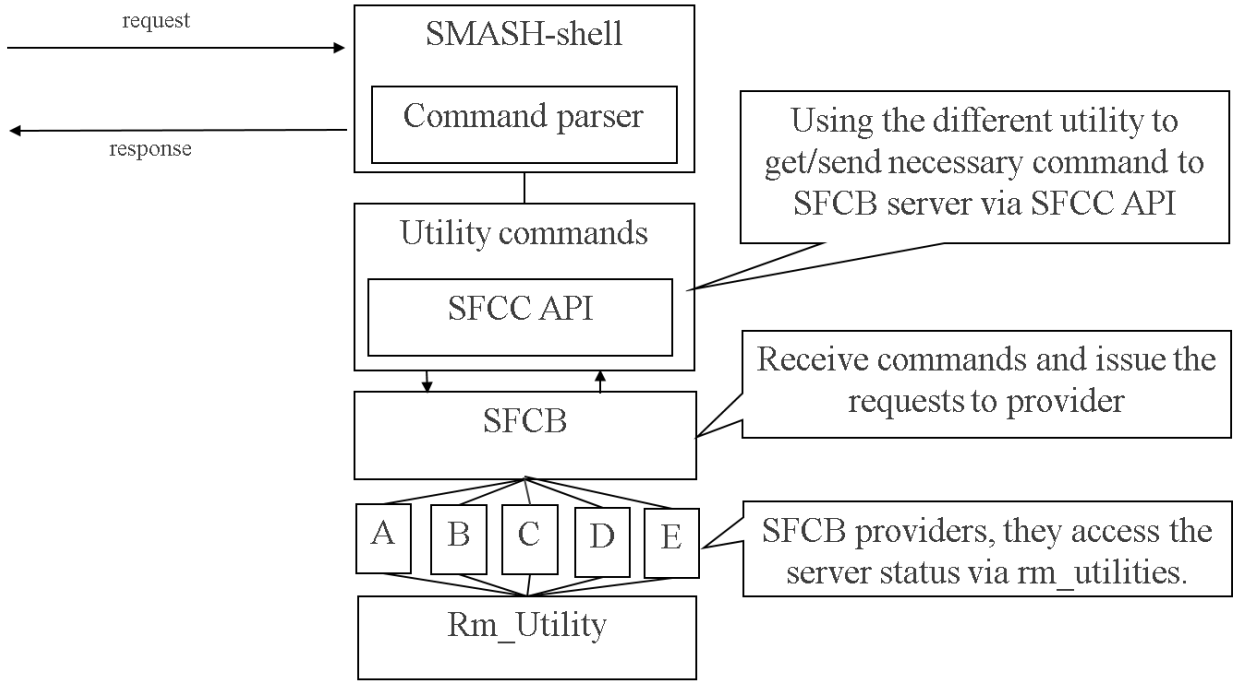


SMASH Block Diagram

1.1 Features

- Allow manager to access and control via the command-line protocol (Support SSH only)
- Support Power control of the server
- Support SM CLP-to-CIM Common Mapping
- SMASH command support

1.2 Flow Chart



The SMASH-shell architecture

1.3 How To Use

SMASH service uses TCP port 2727, any ssh client can connect to the service.

```
robert@Robert-Shih: ~
robert@Robert-Shih:~
$ ssh -p 2727 -l root 192.168.77.200
The authenticity of host '[192.168.77.200]:2727 ([192.168.77.200]:2727)' can't b
e established.
RSA key fingerprint is SHA256:rbHOD00pt+dhLLy5s07saEdYrz9BVIMDPIzig2Zdscc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.77.200]:2727' (RSA) to the list of known ho
sts.
root@192.168.77.200's password:

SMASH-CLP System Management Shell

->
```

1.4 Examples

Following sections are the examples of the output

- **BMC firmware version**

```
-> show
/system1/firmware1

  Targets :
        none

  Properties :
        InstanceID=IPMI:BMC 0xb009bf22
        TargetOSTypes=73
        TargetOSTypes=94
        TargetOSTypes=100
        TargetOSTypes=106
        OtherExtendedResourceTypeDescription=Binary File
        ExtendedResourceType=1
        IsEntity=true
        IdentityInfoType=CIM:SoftwareFamily
        IdentityInfoValue=Firmware:BMC
        TargetTypes=BMC
        SerialNumber=............
        Classifications=10
        Languages=eng
        Manufacturer=Insyde Software Corp.
        IsLargeBuildNumber=false
        MinorVersion=24
        MajorVersion=35
        BMC FW Rev=35
        Build ID=44.923700

  Verbs :
        cd
        show
        help
        version
        exit
```

- **Get SEL Log**

```
-> show
/system1/logs1/log1

  Targets :
        record001
        record002

  Properties :
        InstanceID=IPMI:IPMI BMC SEL Log
        Name=IPMI SEL
        TransitioningToState=12
        EnabledDefault=2
        RequestedState=12
        EnabledState=1
        LogState=4
        OverwritePolicy=0
        CurrentNumberOfRecords=2

  Verbs :
        cd
        show
        help
        version
        exit
        delete
```

- **Get SEL Entry**

```
-> show record001
/system1/logs1/log1/record001

  Targets :
        none

  Properties :
        RecordID=1
        CreationClassName=CIM_LogRecord
        LogName=IPMI SEL
        LogCreationClassName=CIM_RecordLog
        Description=Log Area Reset/Cleared - Asserted
        Time Stamp=Thu Nov 28 02:50:55 2024
        Controller=BMC
        Sensor Type=Event Logging Disabled
        Sensor Name=System_Event_Log
        Event Severity=OK

  Verbs :
        show
        help
        version
        exit
```

- **Power Control**

```
-> show
/system1/pwrmgtsvc1

  Targets :
        none

  Properties :
        Name=IPMI Power Service
        CreationClassName=IPMI_PowerManagementService
        ElementName=Server Power Management Service
        TransitioningToState=12
        EnabledDefault=2
        RequestedState=12
        EnabledState=5
        PowerState=0

  Verbs :
        cd
        show
        help
        version
        exit
        start
        stop
        reset
```

- **Sensor Information**

```
-> show
/system1/sensors1

  Targets :
        sensor001
        sensor002

  Properties :
        TransitioningToState=12
        EnabledDefault=2
        RequestedState=12
        EnabledState=5
        CurrentPowerStatus=OFF

  Verbs :
        cd
        show
        help
        version
        exit
-> show sensor001
/system1/sensors1/sensor001

  Targets :
        none

  Properties :
        DeviceID=2.89.0.32.6F.0
        CreationClassName=Discrete_Sensor
        SystemName=IPMI BMC
        SystemCreationClassName=ComputerSystem
        Description=BMC_ECC_Err(89.0.32):46
        Caption=Memory(89.0.32)
        Name=CIM_Sensor
        TransitioningToState=12
        EnabledDefault=2
        RequestedState=12
        EnabledState=5
        CurrentState=Normal
        SensorType=12

  Verbs :
        show
        help
        version
        exit
```